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Parameter Identification of Lithium-Ion Batteries Model to Predict Discharge Behaviors Using Heuristic Algorithm

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A complete set of parameters that describes lithium-ion batteries is required in order to better control the sophisticated battery management system used during practical application of lithium-ion batteries. Previous work in the literature requires long times and a computer cluster to identify the full set of parameters while using a pseudo-two-dimensional (P2D) model and a genetic algorithm (GA). This work proposes a more efficient parameter identification method based on the heuristic algorithm. The heuristic algorithm includes features such as the requirement of fewer identified parameters and the effective interval spacing of parameters to save searching time during the GA. This algorithm also uses a divide-and-conquer strategy that divides the full set of P2D parameters into two groups and identifies each group, respectively. With this method, the complete identification of P2D model parameters can be performed within 10 h using a single core computer. Simulated data using the above identified parameters are consistent with the reference results from the literature. And the identified paramters could be used for estimation of the state of charge and the state of available power capability of lithium-ion batteries.
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In recent years, there has been a rapid development of new energy sources due to the depletion of fossil fuels and increased levels of environmental concern. Because of the time-dependent nature of these new energies (i.e. wind and solar power),[1] energy storage systems are often introduced into the power system to maintain a stable power supply output.[2,3] Lithium-ion batteries (LIBs) have been widely employed in energy storage because of their relatively high power levels and energy densities. However, due to material and structural problems, LIBs are prone to overshooting, overcharging, overheating, and degradation during the practical application.[4,5] Therefore, the inner states of LIBs need to be fine-tuned by precise estimation with particular attention to charge build up and the power capacity.[6–9]

The inner states of the LIBs have been estimated by simulation analysis[10] based on equivalent circuit models,[11–14] electrochemical impedance spectroscopy (EIS) models[15–17] and electrochemical models.[18–25] Because electrochemical models have the ability to reflect the intrinsic mechanism of LIBs from particle reactions, they are considered the best way to precisely estimate the state of LIBs.[21,22] Among various electrochemical models, the pseudo-two-dimensional (P2D) model[18,19,21,24–40] would provide more fundamental understanding of the internal transport phenomena of LIBs. The P2D model consists of a set of nonlinear partial differential equations based on the first principle calculation. If all parameters used in the P2D model were known, the states of LIBs could be accurately determined under any set of conditions.[27] Unfortunately, some parameters cannot be precisely measured. Furthermore, with an increase in change/discharge cycle number, the capacity will gradually decay[41] and the internal state of the battery will become changed. Therefore, in order to safely and accurately control the lithium batteries during their life cycle, all parameters of the lithium battery must be identified. However, the identification of more than 17 parameters still remains a big challenge.[19,21]

Many methods are currently used to estimate the parameters of LIBs, including the Kalman filter method,[6,7,9–12,42,43] recursive least squares methods (including the gradient descent, Levenberg-Marquardt and Gauss-Newton methods),[10,11,13,44–46] non-recursive least squares methods,[10] evolutionary computation methods,[9,14,15,18,21] and computational learning theory.[10] Among these methods, the recursive least squares methods and the evolutionary computation methods have been used to identify the parameters based on the electrochemical models. For example, Boovaragavan et al. used Gauss-Newton method to estimate three parameters based on the reformulated P2D model.[45] Santhanagopalan et al. used Levenberg-Marquardt method based on the single particle model to identify five parameters of lithium-ion batteries.[46] If we adopt the simplest gradient descent method to identify the parameters of P2D models, two problems need to be addressed: (1) Initial value of θ: It is difficult to find a suitable initial value of θ. In this case, the simulated error would be large. (2) Step size: Although an adaptive step size is adopted, it is very difficult to select a suitable step size when the magnitude difference of θ_i is large. Otherwise, the obtained parameters will present a big error and the identification process will take a long time. The technically evolutionary computation method belongs to the family of trial and error problem solvers. This method is a search heuristic that mimics the process of natural selection with a stochastic optimizations character. This method is mostly applied for black box problems especially in the field of expensive optimization. Its main advantage over other algorithms is that it can identify a large number of parameters. For example, Forman et al.[21] divided the model parameters into identifiable and unidentifiable parameters using the Fisher information matrix. The identifiable parameters were estimated from experimental data. They used the P2D model to determine the identifiable parameters. It took three weeks to complete an 88 parameters identification using a cluster of five quad-core computers and a genetic algorithm. The accuracy of the resultant full parameter set was then assessed using the Fisher information and it was found that parameter identifiability was a function of which parameter was being identified and which parameters were already assumed known. Identifiability also depended on the values of the parameters after they had been fit to the data. Zhang et al.[19] established a parameter sensitivity matrix and divided the parameters into different groups according to their average sensitivities. The results of the parameter sensitivity analysis were only valid for a given set of benchmark values and a stepwise experiment was performed to identify the parameters of a multi-physical model using a cluster with 20 cores to predict both the terminal voltage and the surface temperature of the LIBs. Zhang et al.[18] used the modified multi-objective genetic algorithm NSGA-II to establish the Pareto fronts and they used the multiple criteria decision making method TOPSIS to select the final identification results. The process took roughly 19 hours to identify the parameters of the thermal-electrochemical model on a computing cluster with 20 cores based on the parameter sensitivity matrix. The drawbacks of the methods used above are that they require a computer cluster to run the algorithms to identify the parameters of the P2D model and they require long computing times to obtain final results. In order to decrease the time required to obtain a full set of parameters for LIBs, this work proposes a heuristic algorithm that decreases the number of identified parameters required and decreases the effective interval spacing of the parameters to save on searching time during GA. A divide-and-conquer strategy is also applied which divides the full

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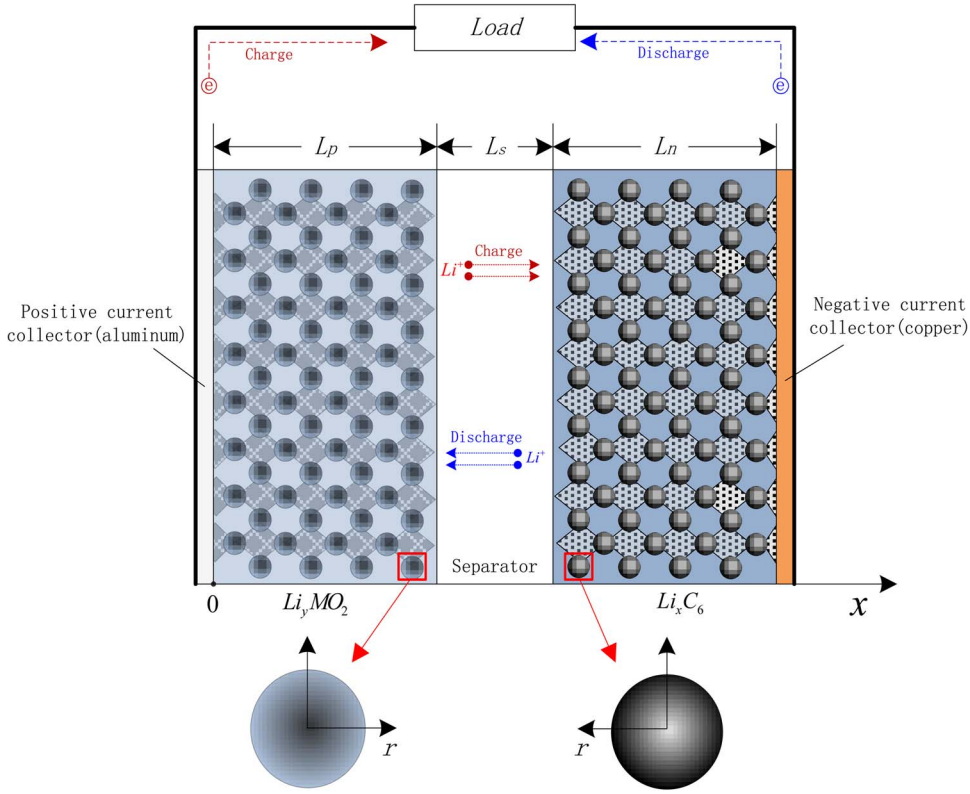

Figure 1. Schematic of P2D model for a typical LIB.

set of P2D parameters into two groups and separately identifies each group.

In this report, the P2D model and the data taken from the literature used for parameter identification are described. Then, the identification process of the P2D model using a heuristic algorithm is depicted. Finally, the identification results of the P2D model and the validation of the effectiveness and rationality are described.

Model Formulation and Data from the Literature

Model formulation.—A schematic of the P2D model is shown in Fig. 1. This model is composed of a negative current collector, a negative electrode, a separator, a positive electrode and a positive current collector. Together, they form a typical "sandwich" structure. There are two interfaces between the electrodes and the separator. The negative/positive electrode and the separator are porous structures. When the lithium-ion battery is fully charged, the active material of anode is full of many Li^+ions. During discharge, lithium ions move from the negative electrode to the positive electrode through the non-aqueous electrolyte and the separator. At the same time, electrons flow through a closed external circuit to perform useful work. When the battery is charging, the process is the same but in reverse. The partial differential equations and algebraic equations that describe the physical and chemical processes of LIB during discharge or charge are shown in Table I and battery parameters are listed in Table II. A more detailed description of the P2D model has been previously reported.[24,25]

Data from the literature.—$LiMnO_4/LiC_6$ battery was used in this work. The battery module of the COMSOL Multiphysics software[27,47] (based on finite element method) was used to calculate

Table I. Governing equations and boundary conditions.

Region	Governing equations*
Solid phase	$\sigma_{eff,j}\frac{\partial^2\phi_{1,j}}{\partial x^2} = a_j F J_j$ $\frac{\partial c_{s,j}}{\partial t}(r_j,t) = \frac{D_{s,j}}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial c_{s,j}}{\partial r}(r,t)\right)$
Electrolyte phase	$\varepsilon_j\frac{\partial c_j}{t}(x,t) = \frac{\partial}{\partial x}\left(D_{eff,j}\frac{\partial c_j}{t}(x,t)\right) + \frac{1-t^+}{F}J_j(x,t)$ $-\sigma_{eff,j}\frac{\partial\phi_{1,j}}{\partial x} - \kappa_{eff,j}\frac{\partial\phi_{2,j}}{\partial x} + \frac{2\kappa_{eff,j}RT}{F}(1-t^+)\frac{\partial \ln c_j}{\partial x} = I$
Electrode-separator interface	$J_j(x,t) = a_j i_{0,j}\left[\exp\left(\frac{\alpha_{a,j}F}{RT}\eta_j(x,t)\right) - \exp\left(\frac{\alpha_{c,j}F}{RT}\eta_j(x,t)\right)\right]$ $i_{0,j} = k_j(c - c_{s,j}^{sur})^{\alpha_{a,j}}(c_{s,j}^{sur})^{\alpha_{c,j}}(c_j)^{\alpha_{a,j}}$ $D_{s,j}\frac{\partial c_{s,j}}{\partial r}\|_{r=R_j} = J_j\frac{\partial c_{s,j}}{\partial r}\|_{r=0} = 0$ $-\sigma_{eff,j}\frac{\partial\phi_{1,j}}{\partial x} = 0, \phi_{2,j}(continue), c_j(continue)$
Electrode-collector interface	$-\sigma_{eff,j}\frac{\partial\phi_{1,j}}{\partial x} = I, -\kappa_{eff,j}\frac{\partial\phi_{2,j}}{\partial x} = 0, -D_{eff,j}\frac{\partial c_j}{\partial x} = 0, \phi_2\|_{x=L_p+L_s+L_n} = 0$

*Positive when $j = p$, negative when $j = n$ and separator when $j = s$.

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Table II. Battery parameters.

Nomenclature*	Unit	Description[24,25]
R_j	m	radius of electrode particle
ε_j	-	porosity of region
$\varepsilon_{l,j}$	-	solution-phase volume fraction of region
c_j	mol m^{-3}	electrolyte concentration
$c_{s,j}$	mol m^{-3}	solid-phase concentration of lithium-ion in electrodes
$c_{s0,j}$	mol m^{-3}	initial solid-phase concentration of lithium-ion in electrodes
$c_{s\,\max,j}$	mol m^{-3}	Max. solid-phase concentration of lithium-ion in electrodes
t^+	-	cationic transport number
D_j	m^2 s^{-1}	salt diffusivity coefficient in electrolyte
$D_{s,j}$	m^2 s^{-1}	Li-ion diffusivity coefficient in the solid electrode particles
σ_j	S m^{-1}	electronic conductivity in the solid-phase electrode
k_j	m$^{2.5}$mol$^{-0.5}$s^{-1}	electrochemical reaction rate constant
$\alpha_{a,j}$		anodic transfer coefficient
$\alpha_{c,j}$		cathodic transfer coefficient
κ_j	S m^{-1}	conductivity of an electrolyte
$\phi_{1,j}$	V	electrical potential in the solid phase
$\phi_{2,j}$	V	electrical potential in the electrolyte
a_j	m^2 m^{-3}	specific interfacial area
F	C mol^{-1}	Faraday's constant
T	K	Temperature
r	m	radial coordinate
x		spatial coordinate or stoichiometry reference point of positive electrode for SOC
y		stoichiometry reference point of negative electrode for SOC
J_j	mol m^{-2}s^{-1}	pore wall flux of lithium ions
I	A m^{-2}	applied current density
$i_{0,j}$	A m^{-2}	exchange current density
L_j	m	Thickness
$Acell$	m^2	cross-section area of the battery

*Positive when $j = p$, negative when $j = n$ and separator when $j = s$.

the P2D model and to obtain six voltage curves of the battery with discharge rates of 0.01C, 0.1C, 0.5C, 1C, 2C and 3C at room temperature and a voltage curve under dynamical load. The parameter values used in the model were taken from the literature.[24,25,48] The discharge curve with rate 0.01C was used to reduce the number of the identified parameters and to identify the physical parameters of the P2D model. The discharge curve with rate 3C was used to identify the dynamic parameters of the P2D model. The other curves were used to validate the identified results. The reason for using these literature parameter values is so that the identified values of the parameters can be directly compared with the corresponding values in the literature after our identification process. In short, the parameter values from the literature are called "true values" and the six voltage curves are called "reference voltage curves".

Heuristic Algorithm

In order to quickly identify the full set of the parameters for the P2D model, the heuristic algorithm was divided into three steps during the parameter identification process. First, the number of identified parameters of the P2D model was decreased. Then, effective intervals times were chosen for identified parameters in order to save searching time during the GA. The last step was to identify the full set of P2D parameters using a divide-and-conquer strategy. Fig. 2 shows the scheme of heuristic algorithm to identify all battery model parameters.

Decrease of the number of identified parameters.—The polarization of the positive and the negative electrode is very small at a discharge current of 0.01C and so it can be ignored. Therefore, the discharge curve at a rate of 0.01C was directly gained from the open-circuit potential curves of the positive and the negative electrode using a curve stretching and superposition transformation and backtracking method.[49]

When the Li-ion battery is fully charged, it is in 100% SOC state and the concentration of Li-ions within the positive electrode is defined as $c_{s0,p}$. The stoichiometric point of positive electrode at 100% SOC is x_0 and the potential of the positive electrode is $U_+(x_0)$ (vs. Li), as shown in Fig. 3. At the same time, the concentration of Li-ions within negative electrode is $c_{s0,n}$ at 100% SOC. The stoichiometric point of negative electrode is y_0 and the potential of the negative electrode is $U_-(y_0)$ (vs. Li). When the Li-ion battery is fully discharged, it is in 0% SOC state and the stoichiometric points of two electrodes are x_1, y_1, respectively.

The value of $|y_0 - y_1|$ was usually not equal to the value of $|x_1 - x_0|$ when the LIB was fully discharged at a rate of 0.01C and the corresponding OCP curves could not be added directly to get the discharge curve of LIB at 0.01C. By curve stretching transformation, $|y_0' - y_1'|$ was equal to $|x_1' - x_0'|$ and we directly gained the discharge curve of the LIB at a rate of 0.01C from corresponding OCP curve using superposition.

In order to identify stoichiometric reference points for x_0, x_1, y_0, y_1, an optimization objective was chosen to minimize the parameter L_0^2 between the reference voltage $V(t)$ and the P2D-simulated voltage trajectories $\vec{V}(t; x_0, x_1, y_0, y_1)$ at a discharge rate of 0.01C. The optimization objective can be expressed as:

$$\begin{aligned} L_0^2: \quad & \min_{\substack{x_0,x_1,\\ y_0,y_1}} \int_0^T \left(V(t) - \vec{V}(t; x_0, x_1, y_0, y_1)\right)^2 dt \\ & s.t. \\ & 0 \le x_0 < x_1 \le 1.0 \\ & 0 \le y_1 < y_0 \le 1.0 \end{aligned} \quad [1]$$

The backtracking algorithm was used to find the minimum value of Equation 1 under constraints and get the optimal values of x_0, x_1, y_0, y_1. The relationships between the stoichiometric reference pointx_0,$y_0$and the solid-phase concentration of lithium ion $c_{s\,\max}$,c_{s0} are as follows:

$$\begin{aligned} x_0 &= \frac{c_{s0,p}}{c_{s\,\max,p}} \\ y_0 &= \frac{c_{s0,n}}{c_{s\,\max,n}} \end{aligned} \quad [2]$$

The maximum solid-phase concentration of lithium ion in the electrodes is an inherent characteristic of the material. Therefore, the reference values in the literature were directly used as the initial values in the GA.[19,50]

The discharge capacity of lithium-ion batteries at very low discharge rate can provide the estimation about electrode volume and stoichiometric cycling range. The low-rate capacity of the negative and positive electrode[50] can be expressed as

$$\begin{aligned} Q_n &= \varepsilon_{s,n}\left(L_n Acell\right)\left(C_{s\,\max,n}\right)(y_0 - y_1)\,F \\ Q_p &= \varepsilon_{s,p}\left(L_p Acell\right)\left(C_{s\,\max,p}\right)(x_1 - x_0)\,F \end{aligned} \quad [3]$$

Thus, the relationship between $\varepsilon_{s,n}$ and $\varepsilon_{s,p}$ can be expressed by:

$$\varepsilon_{s,n} = \frac{\varepsilon_{s,p}\left(L_p\right)\left(C_{s\,\max,p}\right)(|x_1 - x_0|)}{(L_n)\left(C_{s\,\max,n}\right)(|y_0 - y_1|)} \quad [4]$$

Effective intervals.—It is observed that the discharge curves often increase or decrease with time when one parameter is changed, while the remaining parameters are fixed. Fig. 4 shows the discharge curves

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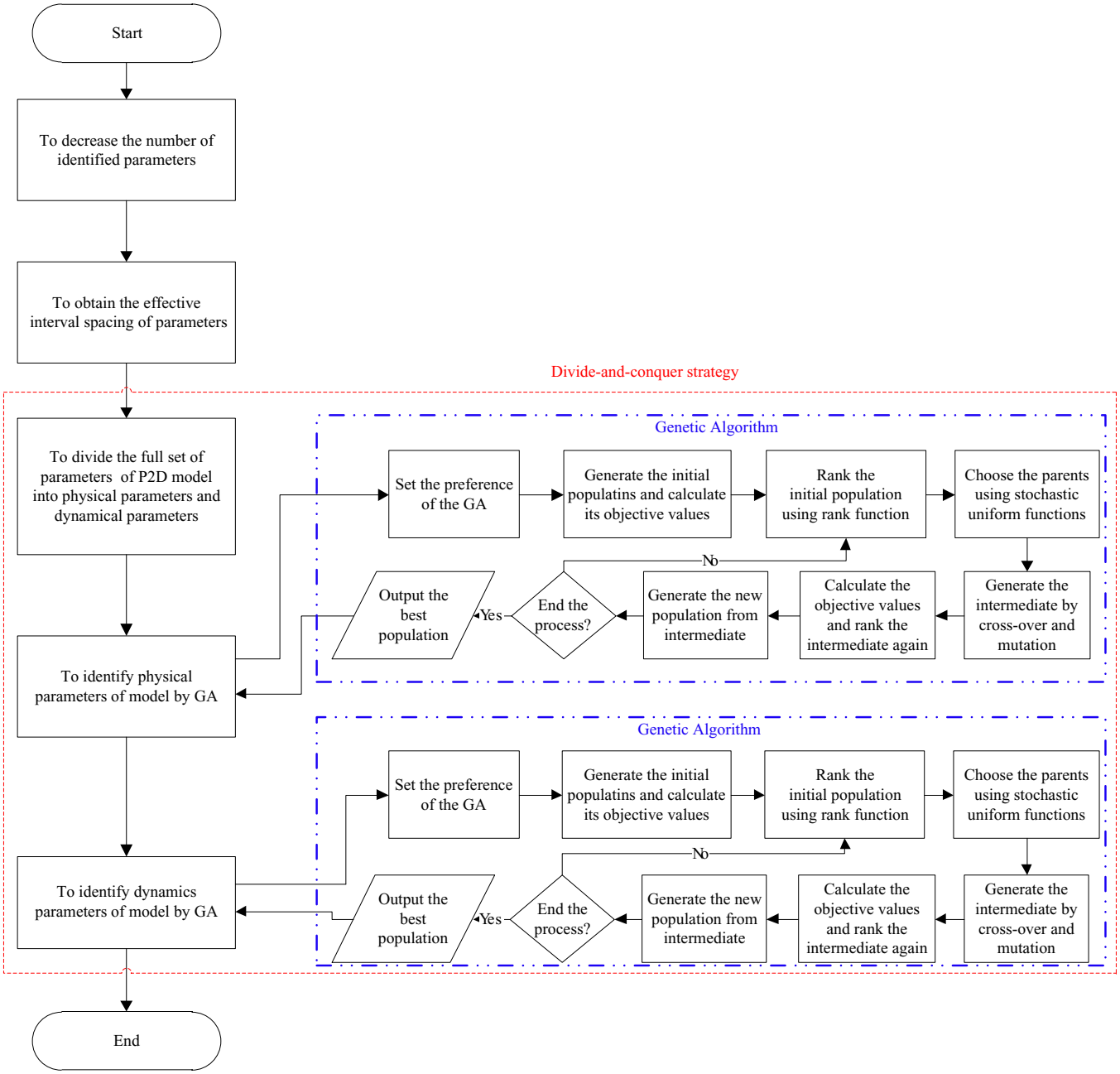

Figure 2. Scheme of heuristic algorithm to identify all battery model parameters.

when the solution-phase volume fractions $\varepsilon_{s,n}$ are 0.3, 0.5 and 0.7. Evidently, the increase in the solution-phase volume fraction shifts the discharge curve from right to left and decreases the battery capacity. If the true value of $\varepsilon_{s,n}$ is 0.5, the discharge curve at $\varepsilon_{s,n} = 0.5$ is used as the reference curve. Other two curves are the simulated curves at $\varepsilon_{s,n} = 0.3$ and $\varepsilon_{s,n} = 0.7$, respectively. Because the reference curve is in between two curves at $\varepsilon_{s,n} = 0.3$ and $\varepsilon_{s,n} = 0.7$, thus, the effective interval for $\varepsilon_{s,n}$ would be [0.3, 0.7]. Therefore, using the battery parameter data from the literature and the reference discharge curves, the effective intervals for every identified parameter of P2D model in GA can be obtained.

Divide-and-conquer strategy1: parameter groups.—In order to identify model parameters quickly, the parameters of the lithium-ion battery in this work are divided into two groups: physical parameters and dynamic parameters. The physical parameters include the thickness, the solid-phase and solution-phase volume fractions, the particle radius of the active materials of the positive and negative electrodes, the thickness of the separator and the cross-section area of the battery. The dynamic parameters include the Li-ion diffusivity coefficient within the electrolyte, the Li-ion diffusivity coefficient in the solid particles of the negative and the positive electrodes, the electrochemical reaction rate constants for the cathode and the anode, and the cationic transport number.

The parameters that are closely related to the actual battery were classified as physical parameters. These parameters include the maximum solid-phase concentration of Li-ions in the positive and negative electrodes, the initial SOC of the positive and negative electrodes, and the initial Li-ion concentration within the electrolyte. The electronic conductivity of the positive and negative electrodes each has a significant effect on the IR drop only at high current densities. Therefore, these parameters were classified as dynamic.

In order to effectively identify these parameters, it was assumed that the sum of the solid-phase volume fraction of the

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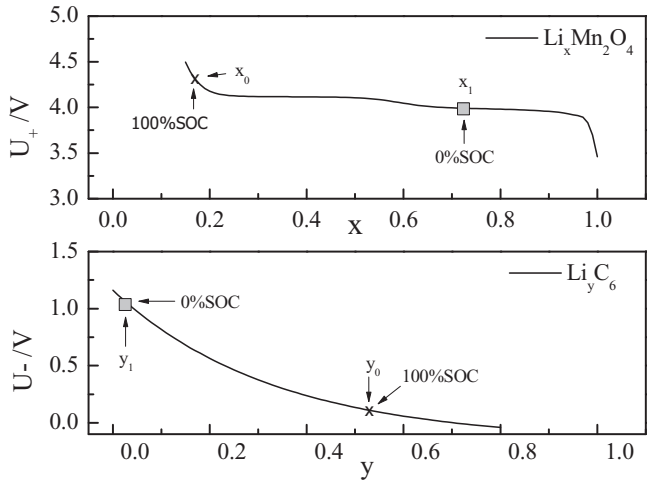

Figure 3. Variation of open-circuit potentials for both negative and positive electrodes with percentage of Li-ion concentration.

electrode, the solution-phase volume fraction of the electrode and the volume fraction of the filler in the electrodes was strictly limited as one.

Divide-and-conquer strategy2: identification parameters.—To identify the physical and dynamic parameters, an optimization objective was chosen to minimize L^2 between the reference voltage $V(t)$ and the P2D-simulated voltage trajectories $\vec{V}(t;\vec{\theta})$ at discharge currents of 0.01C and 3C with respect to the P2D model parameters $\vec{\theta}$. The optimization objective can be expressed as:

$$L^2\colon \min_{\vec{\theta}} \int_0^T (V(t) - \hat{V}(t;\vec{\theta}))^2 dt \qquad [5]$$

Fig. 5 shows the optimization scheme of physical and dynamic parameters based on the genetic algorithm. The inputs to the P2D model are the current profiles versus time at rates of 0.01C and 3C. We experimentally applied these current profiles to the lithium-ion battery and measured the terminal voltage of the battery. The evolution then began from a population of randomly generated individuals to initialize the GA. Each individual was a P2D model parameter set. The evolution was also an iterative process where the population in each iteration was called a generation. In each generation, the L^2 fitness of every individual in the population was evaluated by simulating the P2D model

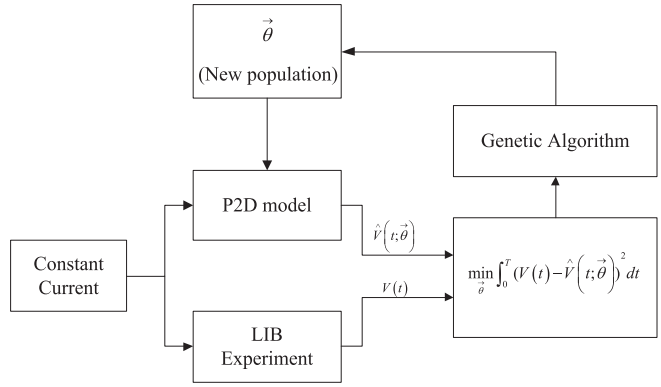

Figure 5. The optimization scheme of physical and dynamics parameters based on the GA.

and compared to the simulated voltage $\hat{V}(t;\vec{\theta})$ versus reference voltage $V(t)$. The most fit individuals were stochastically selected from each population and each individual's genome was modified (e.g. recombined and randomly mutated) to form a new generation. The new generation of candidate solutions was then used in the next iteration of the algorithm. The algorithm terminated when either a maximum number of generations had been produced or a satisfactory fitness level was reached for the population. We observed that with a smaller L^2 fitness value, a better identification result of the parameters was obtained. Additionally, the final values of the parameters were obtained from the fittest member of the population.

The code of the heuristic algorithm is implemented in MATLAB. We set up the population size for 200 and the maximum generation for 100. The P2D model was solved by the battery module of the COMSOL Multiphysics software. With the heuristic algorithm, the full set of P2D parameters can be performed within 10 h using a single core computer. And it takes additional 30 minutes to identify the effective intervals for all parameters.

Results and Discussion

All parameters of the P2D model were successfully identified. The identified values of the parameters are given in Table IV and five discharge curves with rates of 0.1C, 0.5C, 1C, 2C and 3C were calculated according to the identified values in Fig. 6.

Fig. 6 shows the simulated data compared with the reference data with five different discharge rates (0.1C, 0.5C, 1C, 2C and 3C). The

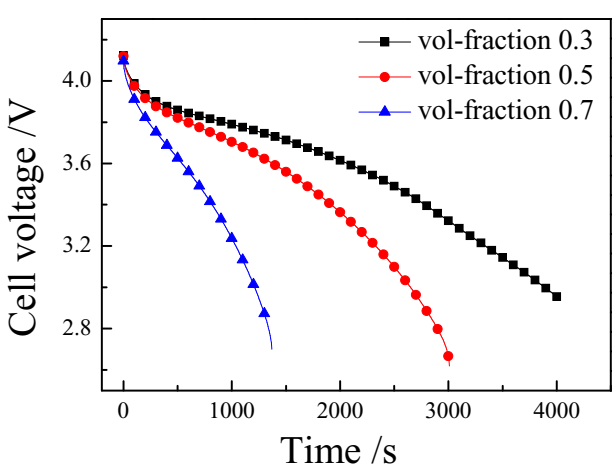

Figure 4. Simulated discharge curves with $\varepsilon_{s,n}$ of 0.3, 0.5 and 0.7, respectively.

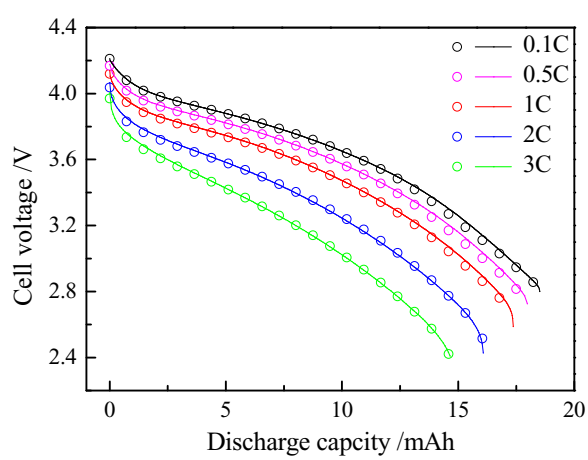

Figure 6. Comparison of simulated data and reference results at discharge rates of 0.1C, 0.5C, 1C, 2C and 3C, respectively. The open circle curves mark the reference results and the solid lines designate the simulated data.

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Table III. Percentile errors and average errors of voltage.

Discharge curves	Percentile error /mV 25%	50%	75%	100%	Average error /mV
0.1C	1.1	3.2	14.5	35.0	9.1
0.5C	3.6	6.3	20.2	37.2	12.9
1C	4.4	8.0	14.2	78.1	11.8
2C	4.1	9.1	11.9	77.4	9.3
3C	2.1	3.5	7.5	34.3	6.4

simulated curves are in good agreement with the reference curves. The reference curves at rates of 0.01C and 3C were used as the objective function to identify the physical and dynamic parameters of the P2D model. The reference curves at rates of 0.1C, 0.5C, 1C and 2C were used to validate the accuracy and robustness of the identification results. The open circle curves mark the reference data and the solid lines designate the simulated data using the identified parameter values. The average errors and percentile errors are shown in Table III. The percentile error of the simulated results with the identified parameters is 78.1 mV or less, and the average error of the simulated results is about 12.9 mV or less.

Figure 7 is a comparison of the simulated capacities and reference capacities at discharge rates of 0.1C, 0.5C, 1C, 2C and 3C, respectively. The black square marker indicates true values and the red dot marker indicates simulated data using identified parameters. There were slight errors when the rate of discharge current was between 0.5C and 2.5C. However, the simulated capacities at low rates and high rates were almost the same as the reference capacities because the discharge curves at rates of 0.01C and 3C were used as the objective curves for the GA. It is also important to note that the simulated capacities at rates of 0.1C, 0.5C, 1C and 2C were predicted values according to the identified parameters. That is to say, if another discharge curve at a rate of 1C or 1.5C was added as the third or fourth objective curve, there would be a better simulated result.

Table V describes the identified stoichiometric reference points of the electrodes at 0% and 100% SOC obtained by the backtracking method. The identified values are almost the same as the true values. The polarization of the positive and the negative electrodes was very small at a discharge current of 0.01C. Therefore, the voltage curve was plotted according to the identified values and the open-circuit potential of the electrodes. Fig. 8 shows a comparison of the simulated curve with the reference curve at a rate of 0.01C. The red line indicates simulated data and the black circle stands for reference data. It is clear that the simulated curve fits well with the reference curve, thus validating our assumption and the identification method of the stoichiometric reference points.

In order to further validate the identified parameters, the simulated data was compared with the reference data under dynamical load. The dynamical load is when the discharge current is not kept constant over

Table IV. Identification results of lithium-ion battery parameters.

Name	Unit	Effective interval	True values[24,25]	Identified values
R_p	m	[5e-6, 1e-5]	8e-6	9.4627e-06
R_n	m	[1e-5, 1.6e-5]	1.25e-5	1.4586e-05
ε_p		[0.3, 0.5]	0.297	0.36337
$\varepsilon_{f,p}$		[0.15, 0.2]	0.259	0.23587
ε_n			0.471	0.477
$\varepsilon_{f,n}$		[0.2, 0.3]	0.172	0.15737
c_0	mol m^{-3}	[1800, 2200]	2000	2084.9648
t^+		[0.1, 0.5]	0.363	0.27851
D	m^2 s^{-1}	[1e-11, 1e-10]	7.5e-11	5.6858e-11
$D_{s,p}$	m^2 s^{-1}	[5e-14, 1.5e-13]	1e-13	0.91892e-13
σ_p	S m^{-1}	[1, 50]	3.8	8.6673
σ_n	S m^{-1}	[90, 200]	100	113.2301
k_n	m$^{2.5}$mol$^{-0.5}$s^{-1}	[1e-11, 3.5e-10]	2e-11	4.3248e-11
k_p	m$^{2.5}$mol$^{-0.5}$s^{-1}	[1e-11, 1.5e-10]	2e-11	2.7934e-11
$c_{s0,n}$	mol m^{-3}	[1e4, 3.5e4]	14870	14941.7
$c_{s0,p}$	mol m^{-3}	[2000, 6500]	3900	3600.4
$c_{s\,\max,n}$	mol m^{-3}		26390	26390
$c_{s\,\max,p}$	mol m^{-3}		22860	22860
L_n	m	[92e-6, 125e-6]	100e-6	97.661e-06
L_p	m	[180e-6, 197e-6]	183e-6	182.6e-06
L_s	m	[50e-6, 66e-6]	52e-6	54.306e-06

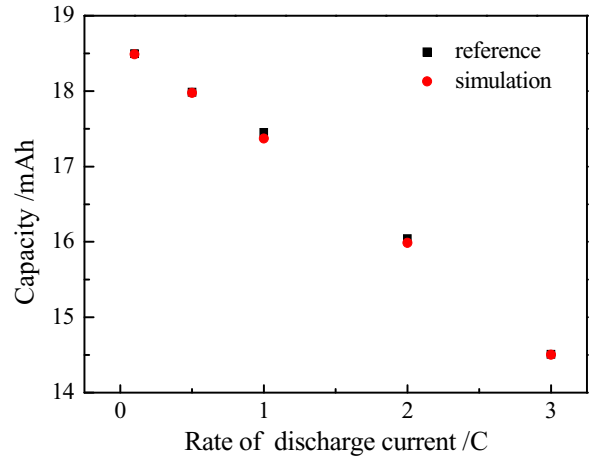

Figure 7. Comparison of simulated capacity with reference one at five different discharge rates.

Table V. Stoichiometric reference points of the electrodes at 0% and 100% SOC.

	True value[24,25]	Identified value
stoichiometry of positive electrode at 100% SOC, x_0	0.1706	0.1706
stoichiometry of positive electrode at 0% SOC, x_1	0.7326	0.7322
stoichiometry of negative electrode at 100% SOC, y_0	0.5635	0.5618
stoichiometry of negative electrode at 0% SOC, y_1	0.0005	0.0

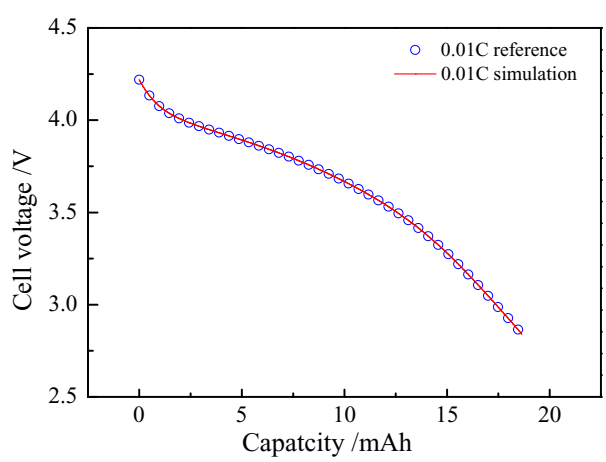

Figure 8. Comparison of simulated data with reference results at a rate of 0.01C.

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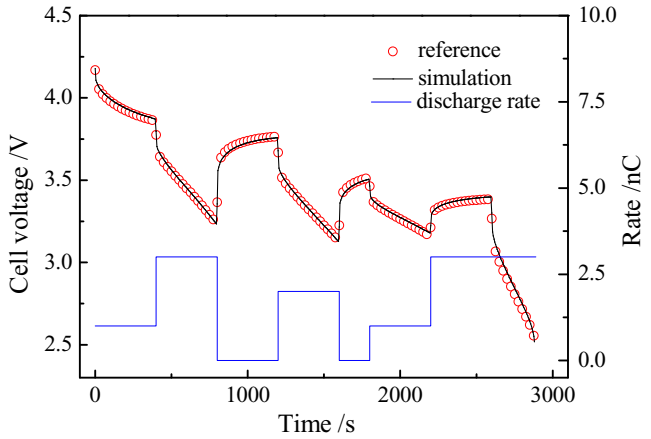

Figure 9. Comparison of simulated curve with reference curve of a LIB under dynamical loads.

the lithium-ion battery discharge period. The simulated and reference curves under dynamical load are shown in Figure 9. The red circle line indicates reference data of the battery voltage and the black solid line stands for the simulated data with identified parameters. The blue line indicates the rate of discharge current. The simulated curve also fits well with the reference curve.

Conclusions

In this work, a more time-efficient parameter identification method for LIBs using a heuristic algorithm was developed. Three steps were used to speed up the identification of the parameters of P2D model. The complete identification of P2D model parameters was performed within 10 h using a single core computer. Simulated results using the identified parameters are in good agreement with the reference results. Furthermore, the identified parameters can be used for estimation of the state of charge and the state of available power capability of LIBs and for modeling the lifetime of LIBs.

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References

1. J. Druitt and W. G. Fruh, *Journal of Power Sources*, **216**, 104 (2012).
2. K. Bradbury, L. Pratson, and D. Patino-Echeverri, *Applied Energy*, **114**, 512 (2014).
3. B. Dunn, H. Kamath, and J. M. Tarascon, *Science*, **334**, 928 (2011).
4. J. Lamb, C. J. Orendorff, L. A. M. Steele, and S. W. Spangler, *Journal of Power Sources*, **283**, 517 (2015).
5. C. Hendricks, N. Williard, S. Mathew, and M. Pecht, *Journal of Power Sources*, **297**, 113 (2015).
6. S. J. Tong, M. P. Klein, and J. W. Park, *Journal of Power Sources*, **293**, 416 (2015).
7. W. G. Zhang, W. Shi, and Z. Y. Ma, *Journal of Power Sources*, **289**, 50 (2015).
8. H. Rahimi-Eichi, F. Baronti, and M. Y. Chow, *Ieee Transactions on Industrial Electronics*, **61**, 2053 (2014).
9. M. R. Khan, G. Mulder, and J. Van Mierlo, *Journal of Power Sources*, **246**, 629 (2014).
10. C. Fleischer, W. Waag, H. M. Heyn, and D. U. Sauer, *Journal of Power Sources*, **260**, 276 (2014).
11. Y. Zou, X. S. Hu, H. M. Ma, and S. E. Li, *Journal of Power Sources*, **273**, 793 (2015).
12. F. C. Sun and R. Xiong, *Journal of Power Sources*, **274**, 582 (2015).
13. T. H. Feng, L. Yang, X. W. Zhao, H. D. Zhang, and J. X. Qiang, *Journal of Power Sources*, **281**, 192 (2015).
14. L. Devarakonda and T. S. Hu, *Journal of Power Sources*, **282**, 19 (2015).
15. S. E. Li, B. J. Wang, H. Peng, and X. S. Hu, *Journal of Power Sources*, **258**, 9 (2014).
16. R. Al Nazer, V. Cattin, P. Granjon, M. Montaru, and M. Ranieri, *Ieee Transactions on Vehicular Technology*, **62**, 2896 (2013).
17. J. L. Leek, A. Chemistruck, and G. L. Plett, *Journal of Power Sources*, **206**, 367 (2012).
18. L. Q. Zhang, L. X. Wang, G. Hinds, C. Lyu, J. Zheng, and J. F. Li, *Journal of Power Sources*, **270**, 367 (2014).
19. L. Q. Zhang, C. Lyu, G. Hinds, L. X. Wang, W. L. Luo, J. Zheng, and K. H. Ma, *J Electrochem Soc*, **161**, A762 (2014).
20. J. Marcicki, M. Canova, A. T. Conlisk, and G. Rizzoni, *Journal of Power Sources*, **237**, 310 (2013).
21. J. C. Forman, S. J. Moura, J. L. Stein, and H. K. Fathy, *Journal of Power Sources*, **210**, 263 (2012).
22. P. Kemper, S. E. Li, and D. Kum, *Journal of Power Sources*, **286**, 510 (2015).
23. X. B. Han, M. G. Ouyang, L. G. Lu, and J. Q. Li, *Journal of Power Sources*, **278**, 814 (2015).
24. T. F. Fuller, M. Doyle, and J. Newman, *J Electrochem Soc*, **141**, 1 (1994).
25. M. Doyle, T. F. Fuller, and J. Newman, *J Electrochem Soc*, **140**, 1526 (1993).
26. P. W. C. Northrop, B. Suthar, V. Ramadesigan, S. Santhanagopalan, R. D. Braatz, and V. R. Subramanian, *Journal of the Electrochemical Society*, **161**, E3149 (2014).
27. L. Cai and R. E. White, *Journal of Power Sources*, **196**, 5985 (2011).
28. L. Q. Zhang, C. Lyu, L. X. Wang, J. Zheng, W. L. Luo, and K. H. Ma, *Advances in Mechanical Engineering* (2013).
29. M. Safari and C. Delacourt, *J Electrochem Soc*, **158**, A562 (2011).
30. P. W. C. Northrop, M. Pathak, D. Rife, S. De, S. Santhanagopalan, and V. R. Subramanian, *Journal of the Electrochemical Society*, **162**, A940 (2015).
31. B. Suthar, P. W. C. Northrop, R. D. Braatz, and V. R. Subramanian, *Journal of the Electrochemical Society*, **161**, F3144 (2014).
32. M. T. Lawder, B. Suthar, P. W. C. Northrop, S. De, C. M. Hoff, O. Leitermann, M. L. Crow, S. Santhanagopalan, and V. R. Subramanian, *Proceedings of the Ieee*, **102**, 1014 (2014).
33. M. T. Lawder, P. W. C. Northrop, and V. R. Subramanian, *Journal of the Electrochemical Society*, **161**, A2099 (2014).
34. B. Suthar, V. Ramadesigan, P. W. C. Northrop, B. Gopaluni, S. Santhanagopalan, R. D. Braatz, and V. R. Subramanian and Ieee, in *2013 american control conference*, p. 5350 (2013).
35. S. De, B. Suthar, D. Rife, G. Sikha, and V. R. Subramanian, *Journal of the Electrochemical Society*, **160**, A1675 (2013).
36. S. De, P. W. C. Northrop, V. Ramadesigan, and V. R. Subramanian, *Journal of Power Sources*, **227**, 161 (2013).
37. V. Ramadesigan, P. W. C. Northrop, S. De, S. Santhanagopalan, R. D. Braatz, and V. R. Subramanian, *Journal of the Electrochemical Society*, **159**, R31 (2012).
38. P. W. C. Northrop, V. Ramadesigan, S. De, and V. R. Subramanian, *Journal of the Electrochemical Society*, **159**, S5 (2012).
39. S. De, P. W. C. Northrop, V. Ramadesigan, R. D. Braatz, and V. R. Subramanian and Ieee, in *2012 american control conference*, p. 4275 (2012).
40. V. Ramadesigan, K. J. Chen, N. A. Burns, V. Boovaragavan, R. D. Braatz, and V. R. Subramanian, *Journal of the Electrochemical Society*, **158**, A1048 (2011).
41. A. Farmann, W. Waag, A. Marongiu, and D. U. Sauer, *Journal of Power Sources*, **281**, 114 (2015).
42. A. Singh, A. Izadian, and S. Anwar, *Journal of Power Sources*, **268**, 459 (2014).
43. B. Balasingam, G. V. Avvari, B. Pattipati, K. R. Pattipati, and Y. Bar-Shalom, *Journal of Power Sources*, **272**, 1142 (2014).
44. C. Fleischer, W. Waag, H. M. Heyn, and D. U. Sauer, *Journal of Power Sources*, **262**, 457 (2014).
45. V. Boovaragavan, S. Harinipriya, and V. R. Subramanian, *Journal of Power Sources*, **183**, 361 (2008).
46. S. Santhanagopalan, Q. Z. Guo, and R. E. White, *J Electrochem Soc*, **154**, A198 (2007).
47. M. Thunman, K. Marquardt, R. Hahn, D. Kober, O. Goerke, and H. Schubert, *Ecs Transactions*, **41**, 147 (2012).
48. M. Doyle, J. Newman, and J. Reimers, *Journal of Power Sources*, **52**, 211 (1994).
49. J. R. Bitner and E. M. Reingold, *Communications of the ACM*, **18**, 651 (1975).
50. K. Smith and C.-Y. Wang, *Journal of Power Sources*, **161**, 628 (2006).

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